

SAURAV G MISHRA

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EDUCATION

Gopalan College of Engineering and Management (VTU) - Bangalore, India
B.E. in Information Science and Engineering | 2025 - 2029 | Year 1 CGPA: 7.72 / 10

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C, C++, SQL, HTML5, CSS3
Web & Backend: Node.js, Express, Flask, Firebase (Auth, Cloud Firestore), SQLite, REST APIs
Automation & Tools: Git, GitHub, Bash, n8n, Zapier, Docker, GitHub Actions (CI/CD)
Systems & Linux: Linux CLI, POSIX Syscalls, Virtual Filesystems (/proc, /sys), QEMU Bring-up

EXPERIENCE

AI Research Intern - Sunny Side Up - May 2025 - Jun 2025 | Bangalore, India

- Benchmarked and evaluated 50+ generative AI tools and APIs across image generation, media workflows, and productivity to establish standard adoption playbooks for the creative team.
- Prototyped automated workflows in Zapier integrating Google Workspace APIs, eliminating manual notification steps across internal routines and onboarding.
- Investigated Photoshop ExtendScript (JavaScript) automation pipelines for programmatic layout adaptation and documented technical integration guides for reproducible deployment.

PROJECTS

FreeDot Linux - Minimal C++ Userspace & Init System | C++20, Linux Syscalls, QEMU, Bash, POSIX

- Explored Linux kernel booting, virtual filesystems (/proc, /sys), and userspace bring-up under a strict 600MB memory budget in QEMU.
- Engineered a custom PID 1 init system in C++20 managing process lifecycle, zombie process reaping (waitpid), signal dispatching, and socket ioctl networking.

StudyForge - Gamified Task & Habit Tracker PWA | JavaScript, HTML5, CSS3, Firebase (Auth, Cloud Firestore)

- Built a client-side productivity PWA with XP progression mechanics, levels, and streak tracking to gamify academic task completion.
- Integrated Firebase Authentication and Cloud Firestore for low-latency state persistence, real-time sync, and client-side data isolation.

CB-watch - Touchless Rubik's Cube Timer | Python, OpenCV, Google MediaPipe

- Engineered an optical speedcubing timer leveraging MediaPipe 21-point hand landmark tracking to detect open-palm gestures without keyboard input.
- Implemented frame-by-frame state machine logic with OpenCV visual cues, preventing false-positive start/stop triggers during inspection.

FoodShare - Surplus Food Redistribution Platform | JavaScript, Node.js, Firebase Firestore, PWA

- Developed a responsive Progressive Web App connecting local communities to donate and claim surplus food, featuring real-time feed updates.

Open Source Contributions (10 Repositories, 7 Merged PRs) | Python, JSON, TypeScript, Markdown, Git, GitHub Actions

- Contributed to 10 open-source codebases across Python, Web, and systems tooling (Lark Parser, Python-Vibe, Composable Data Stack, VS Code Agent Bell, Kana-Dojo, CONTINUUM), adhering to upstream CI and code review standards.